

MORAWA, Australia

WMO: 944170

Lat: 29.2039S

Lon: 116.0247E

Elev: 272

StdP: 98.10

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	4.4	-7.0	2.2	24.1	-4.9	2.6	24.6	10.1	17.8	9.1	16.8	1.4	170	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	40.9	19.5	39.3	19.3	37.8	19.1	23.5	31.1	22.5	31.1	21.6	31.2	4.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	17.0	25.8	20.1	15.3	25.3	18.6	13.9	25.1	71.5	30.9	67.8	31.4	64.2	31.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.8	8.0	0.9	43.8	0.9	1.5	0.3	44.9	-0.2	45.7	-0.7	46.6	-1.3	47.7		
			-0.1	25.3	1.2	1.8	-1.0	26.5	-1.7	27.6	-2.4	28.6	-3.3	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	28.5	28.2	25.6	21.6	17.0	13.7	12.4	13.3	15.2	19.4	23.5	26.0
	DBStd	6.52	3.37	3.35	3.44	3.19	2.82	2.40	2.21	2.41	2.95	3.39	3.71	3.82
	HDD10.0	8	0	0	0	0	0	1	5	2	1	0	0	0
	HDD18.3	683	0	0	1	8	59	139	185	157	103	27	3	1
	CDD10.0	3773	573	508	485	347	217	112	79	104	156	292	404	496
	CDD18.3	1406	314	276	227	104	18	1	0	1	9	60	157	238
	CDH23.3	17976	4291	3614	2628	1007	198	12	3	37	184	845	2009	3149
CDH26.7	10084	2696	2164	1409	407	42	1	0	4	55	354	1092	1860	
Wind	WSAvg	3.9	5.0	4.7	4.5	3.6	3.1	3.1	3.0	3.1	3.6	4.0	4.5	4.8
Precipitation	PrecAvg	331	14	18	21	23	50	58	52	36	24	14	10	9
	PrecMax	520	93	110	220	107	167	176	126	112	67	82	45	36
	PrecMin	172	0	0	0	0	3	8	5	7	5	0	0	0
	PrecStd	83	21	24	39	25	40	31	28	19	15	14	10	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	41.9	39.9	35.6	30.2	24.9	23.2	26.7	31.3	35.9	39.8	41.9
		MCWB	19.3	19.8	18.7	17.7	15.7	13.5	13.2	13.5	15.2	16.4	18.3	19.0
	2%	DB	41.3	39.7	37.7	33.2	27.6	22.9	20.9	23.9	28.2	33.2	37.4	39.9
		MCWB	19.8	19.8	19.0	17.0	14.8	12.6	12.7	13.1	14.3	16.3	17.3	18.8
	5%	DB	39.5	38.2	35.8	31.0	25.5	21.1	19.2	21.5	25.3	30.6	35.3	37.7
		MCWB	19.5	19.8	19.0	16.9	14.4	12.7	12.6	12.8	13.7	15.4	16.7	18.6
	10%	DB	37.6	36.5	33.9	28.9	23.5	19.2	17.7	19.5	22.6	28.1	33.0	35.6
		MCWB	19.5	19.9	18.6	16.6	14.1	12.6	12.4	12.7	13.4	14.9	16.3	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	24.6	23.2	20.7	19.3	17.3	16.9	16.8	17.3	18.8	20.5	22.7
		MCDB	29.5	31.3	28.4	27.4	22.6	19.0	18.2	18.7	23.7	28.2	31.9	32.4
	2%	WB	23.6	23.5	22.0	19.7	17.8	16.0	15.2	15.6	16.2	17.5	19.0	21.1
		MCDB	31.9	30.8	28.6	25.9	21.9	19.0	17.1	18.9	22.7	27.3	31.7	33.3
	5%	WB	22.6	22.6	21.0	18.9	16.7	15.0	14.3	14.7	15.3	16.7	18.2	20.1
		MCDB	31.8	31.0	30.0	26.1	21.2	17.8	16.5	18.4	21.5	26.2	29.7	32.4
	10%	WB	21.6	21.6	20.1	18.0	15.7	14.1	13.4	13.8	14.4	16.0	17.5	19.2
		MCDB	31.4	31.8	29.4	25.1	20.7	17.0	16.2	17.6	20.3	25.1	28.7	32.1
Mean Daily Temperature Range	5% DB	MDBR	16.3	14.9	14.1	13.6	12.8	11.4	11.3	12.7	14.3	15.8	16.1	16.2
		MCDBR	18.7	16.8	16.6	16.4	15.7	14.1	13.9	16.6	18.9	19.4	19.2	18.7
		MCWBR	5.6	4.9	4.9	5.6	6.5	6.8	7.3	7.5	7.4	6.3	5.9	5.7
	5% WB	MCDBR	13.7	13.3	13.2	12.6	10.7	9.4	9.6	11.5	14.4	16.0	16.6	16.3
		MCWBR	4.6	4.6	4.6	4.9	5.5	5.7	6.3	6.4	6.4	5.6	5.3	5.6
Clear-Sky Solar Irradiance	taub	0.335	0.341	0.323	0.310	0.292	0.283	0.282	0.287	0.306	0.312	0.318	0.325	
	taud	2.553	2.554	2.584	2.599	2.617	2.638	2.629	2.607	2.523	2.523	2.532	2.535	
	Ebn at Noon	1006	982	970	935	907	896	911	947	971	1000	1017	1019	
	Edh at Noon	109	106	97	86	76	71	74	84	101	108	111	111	
All-Sky Solar Radiation	RadAvg	8.18	7.40	6.32	4.93	3.84	3.24	3.44	4.37	5.69	7.05	7.99	8.47	
	RadStd	0.47	0.39	0.40	0.24	0.24	0.21	0.19	0.14	0.26	0.30	0.30	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page